AP CALCULUS AB

Unit 1 • Lesson 1-4 • Guided notes

Lesson 1-4 • Continuity at a Point

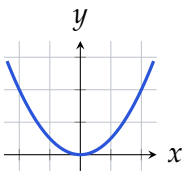
Guided notes • Fill in the blanks and work the examples in the boxes

Objective: identify the types of discontinuities and prove continuity with the continuity test.

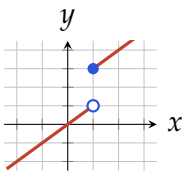
What does it mean to be continuous?

A graph is continuous if you can trace it _____.

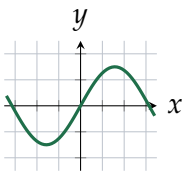
Are these graphs continuous or discontinuous?



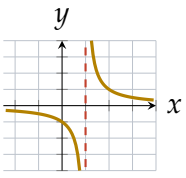
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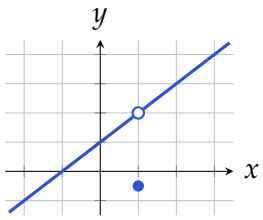


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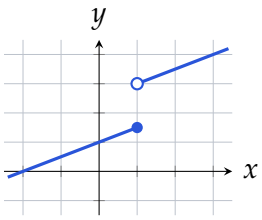
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Types of discontinuities



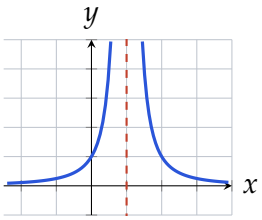
_____

the limit _____ but is not equal to $f(c)$



_____

the one-sided limits are _____



_____

the function goes to _____

An **endpoint** (like the start of $y = \sqrt{x}$ at $x = 0$) only needs the one-sided limit from the _____ of the domain.

Continuity test

f is continuous at $x = c$ when

_____ = _____ = _____

left right actual height

Example 1

$$h(x) = \begin{cases} 2x - 1, & x \leq 3 \\ x^2 - 4, & x > 3 \end{cases}$$

Prove that h is continuous at $x = 3$.

Example 2	For what value of k is g continuous at $x = 1$? $g(x) = \begin{cases} 3x^2 - k, & x \leq 1 \\ kx + 5, & x > 1 \end{cases}$
Summary	